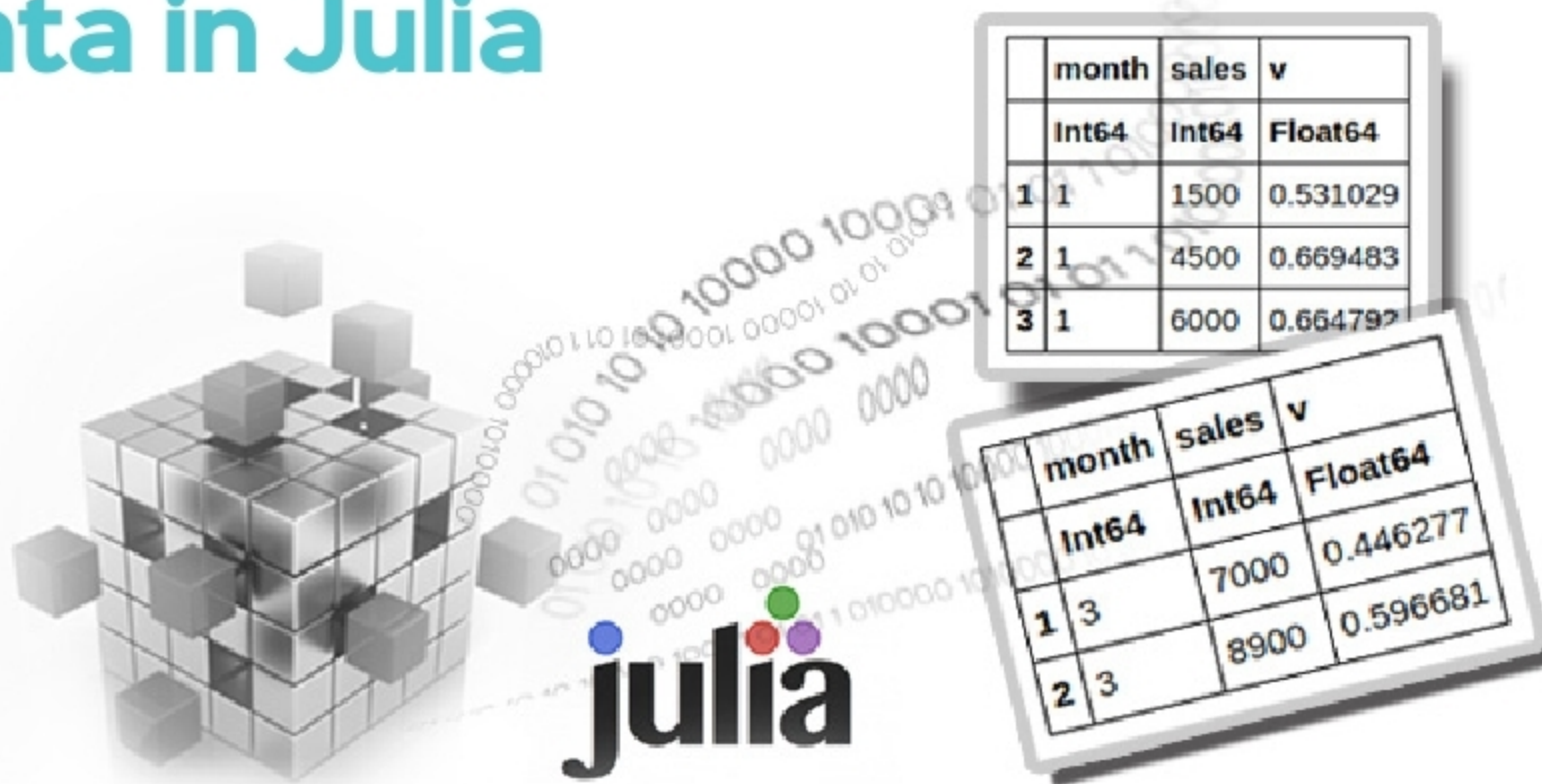


# DataFrames.jl: Handling In-memory Tabular Data in Julia



Effective handling of data is very important for the success of any software application. If a language provides an option to efficiently handle in-memory tabular data, it will make a developer's life simple. This article introduces DataFrames.jl, an excellent package that you can use with Julia to handle in-memory tabular data, and explores its rich set of features.

**D**ata is the lifeline of any software application. The success or failure of an application depends on the efficiency with which the data is represented and handled by it. During the earlier days, we had data structures such as arrays to handle a collection of homogenous data. When it comes to two-dimensional data, there are matrices that are built using two-dimensional arrays. Multi-dimensional data can be handled with multi-dimensional arrays.

Tabular data is one of the common types of data that we frequently encounter in real-time applications. Though we can handle tabular data with plain matrices, there are certain limitations in terms of the operations that we can do directly. To make the in-memory

tabular data handling simpler, Julia has provided a very effective package called DataFrames.jl. This article explores the features of DataFrames.jl, and illustrates the preliminary steps involved in the efficient handling of tabular data using this software.

(If you are new to the Julia programming language, then you might consider reading this article, which introduces Julia, at <https://www.opensourceforu.com/2016/10/julia-language/>. Julia is a powerful language, and it is getting popular in various domains.)

## Getting ready with DataFrames.jl

Any package in Julia is added using *Pkg* and *add*. Similarly, *DataFrames.jl* can also be added with the following commands:

```
using Pkg
Pkg.add("DataFrames")
```

The code shown in this article has been checked with Julia version 1.4.2.

To use DataFrames.jl, just place the following statement at the beginning of your code:

```
using DataFrames
```

## What is a DataFrame?

A DataFrame is used to represent tabular data. It is represented as a series of vectors. These vectors represent columns. The easiest way to construct a DataFrame is to supply the column vectors. A sample is given below:

```
df = DataFrame(A = 1:5, B = ["M", "F", "F", "M", "F"])
```

